01 / EXECUTIVE OVERVIEW

NEBULA

RL-assisted analogue equalizer design | 5 Gbps NRZ | SKY130 130 nm

A specification-to-circuit workflow that combines a learned search policy, physical circuit simulation and deterministic acceptance checks. The final demonstrated product exports one fixed, source-degenerated CTLE with a transistor reference and physical MIM bypass, together with the evidence needed to review it.

FINAL DEMONSTRATION	RESULT
Requested response	9 dB peaking at 1.90 GHz
Nominal measured response	8.834 dB at 1.769 GHz
Fixed-circuit verification	45/45 PVT points; 315/315 electrical-model cases
Worst modelled eye	133.52 mV height; 0.7656 UI width
Worst simulated noise / power	0.768 mVrms / 9.802 mW (CTLE + bias)

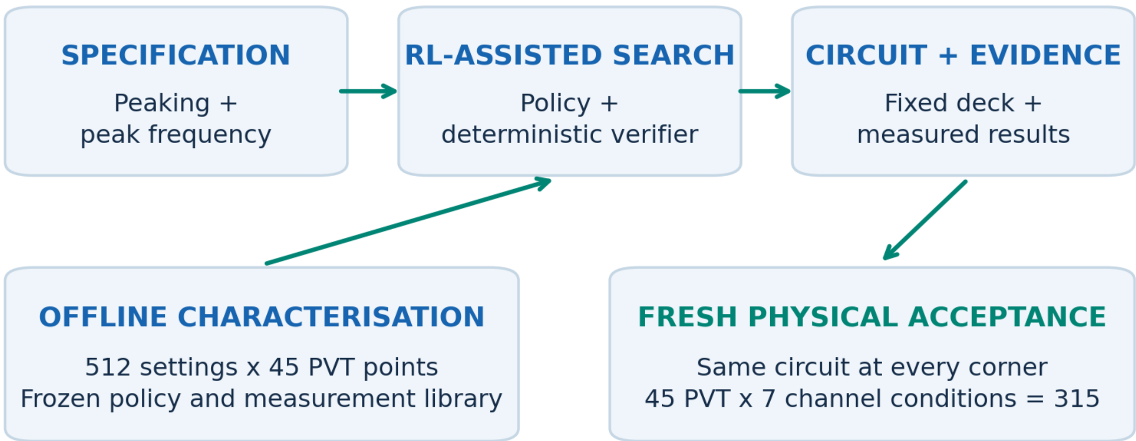

Figure 1. The product separates fast proposal generation from fresh physical acceptance. All headline values come from the saved physical-product run [E1].

Submission contribution

An implemented automation product with a verified physical CTLE example, repeatable RL evidence and inspectable exports. The 1-tap DFE is behavioural; full receiver layout, area and power closure are outside the demonstrated hardware boundary.

02 / REQUIREMENTS

From brief to acceptance

The competition asks for a working equalizer, not eye opening in isolation.

BRIEF REQUIREMENT	IMPLEMENTED INTERPRETATION / EVIDENCE
5 Gbps NRZ; 2.5 GHz Nyquist	Fixed link rate; differential signal convention throughout.
Peaking 3-12 dB; tunable peak 1.25-2.5 GHz	User inputs are peaking and peak frequency. Final physical demonstration: 9 dB / 1.9 GHz.
Source-degenerated CTLE + 1-tap DFE	SKY130 transistor CTLE; physical Rs/Cs geometry selected by software; behavioural first-post-cursor cancellation.
HD3 < -30 dB at 100 MHz, 100 mV differential input	SPICE tone test uses 100 mV differential peak. A separate 2.5 GHz test checks high-frequency linearity.
Input-referred noise < 1.5 mVrms	Differential input reference; 10 MHz-5 GHz integrated SPICE noise.
Power < 15 mW; area < 0.05 mm2	Supply-current power measured for CTLE + reference. Geometry inventoried; complete receiver area/power require integration.
Eye > 100 mV and > 0.4 UI	Worst-ISI cursor model; absolute differential voltage and contiguous positive-opening width.
TT / SS / FF / SF / FS; VDD +/-5%; 0-125 C	45 sampled combinations: 5 processes x 1.71/1.80/1.89 V x 0/27/125 C. Same circuit at every point.
Automated Python flow + simulator + schematic	CLI and browser product; frozen RL policy, verifier, ngspice deck, readable schematic and evidence export.

Declared acceptance assumptions. Request matching uses +/-1.5 dB and +/-0.3 octave, in addition to the absolute response band. These are project tolerances, not extra numbers supplied by the competition. Channel loss is a verification condition, not a CTLE gain request.

PVT scope. Transistor process corners are combined with typical passive models. Independent resistor/capacitor spread, local mismatch and extracted layout parasitics are separate closure tasks. The grid samples the temperature range at three points; it is not a continuous-temperature guarantee.

How to read the result

Electrical acceptance and physical integration are reported separately: the 315-case grid verifies circuit/link behaviour under the stated conditions; layout and complete receiver area remain integration milestones.

03 / AUTOMATION ARCHITECTURE

Fast proposals. Physical decisions.

The final flow makes the boundary between learning, search and circuit verification explicit.

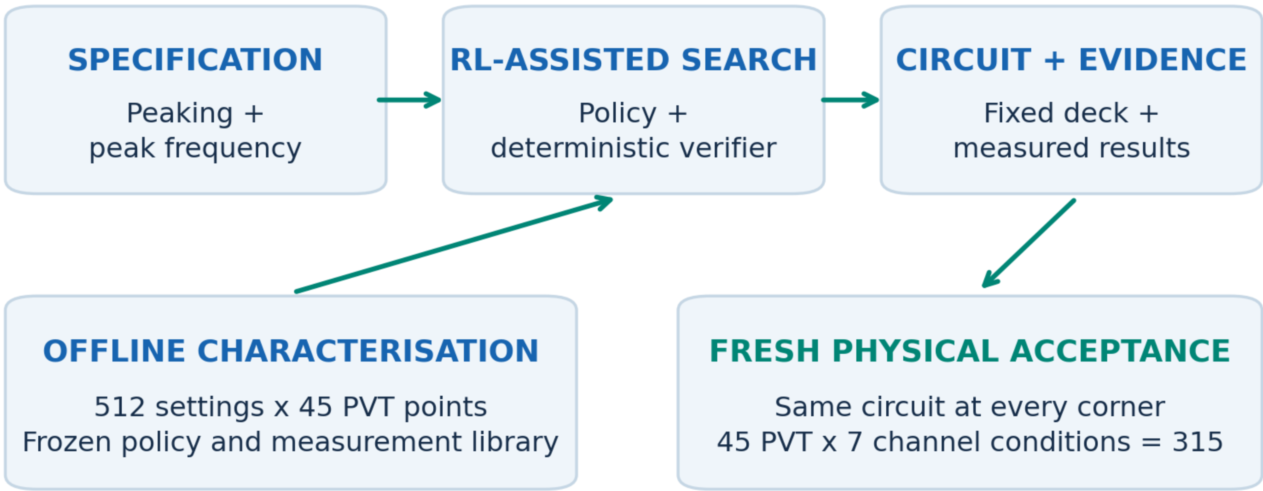

Figure 2. Offline data are reused for proposals. Acceptance of the physical-bias export requires fresh simulator evidence, not a cached legacy score.

01

1. Interpret a target

Validate peaking and peak frequency. Natural-language assistance maps a request into these structured fields; engineering constraints remain in the verifier.

02

2. Propose and intersect

The frozen RL-hybrid policy proposes a library setting. A deterministic all-corner intersection identifies fixed setting 490 (A7, Rs index 5, Cs index 2).

03

3. Realise and remeasure

Instantiate the physical reference and MIM bypass. Check DC/AC/noise and two HD3 tones at all 45 PVT points; evaluate seven link conditions per point.

04

4. Export or explain

Export the exact accepted deck, schematic, structured results and raw evidence. An unsupported target receives an explicit status, not a fabricated compliant circuit.

Attribution matters. The deployed result is a hybrid system: RL supplies a proposal, classical verification enforces a fixed-circuit decision, and fresh SPICE establishes the final evidence. The physical-bias circuit was not used to retrain the frozen policy.

04 / CIRCUIT ARCHITECTURE

The source-degenerated CTLE

A differential NMOS pair with resistive loads and a shared Rs || Cs degeneration network.

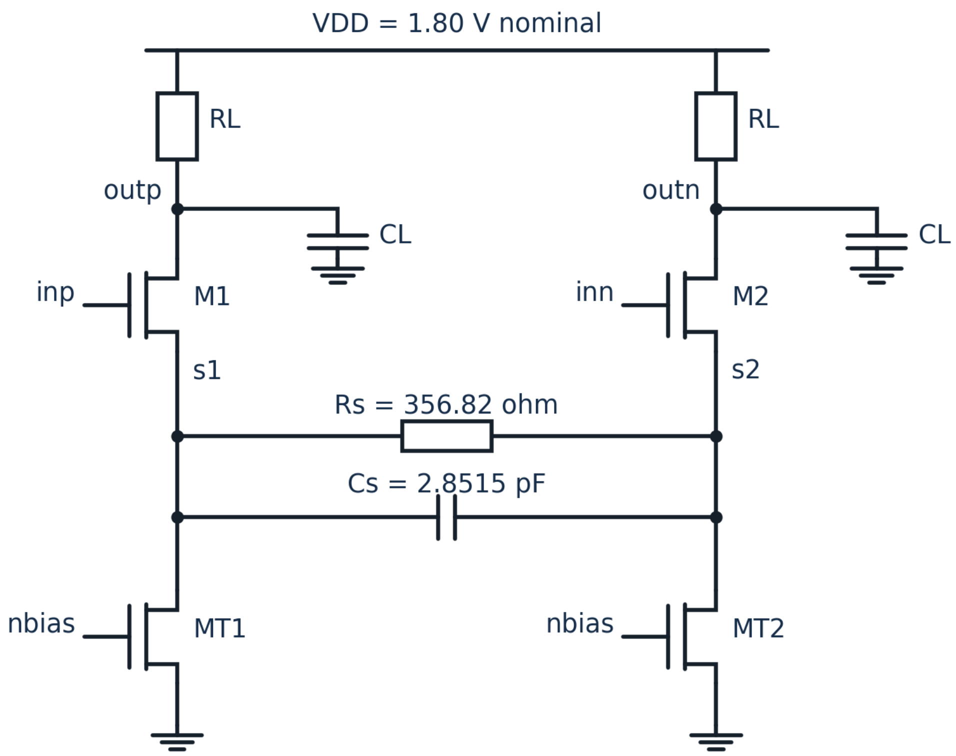

Figure 3. Connectivity redrawn from the exported deck and checked against all physical instances. Input attenuator and bias are expanded on the following pages. NMOS bulks connect to ground.

DEVICE / ELEMENT	FINAL NOMINAL GEOMETRY OR VALUE
M1 / M2	W = 57.9767 um total; L = 0.39115 um; nf = 4
MT1 / MT2	W = 201.656 um; L = 0.5 um; nf = 8
RL per side / external CL	254.633 ohm / 32.628 fF per output
Rs / Cs between sources	356.818 ohm / 2.85155 pF; fixed drawn elements

At low frequency Rs reduces effective transconductance. At high frequency Cs bypasses the degeneration, raising the response before output parasitics impose roll-off. The input attenuator controls signal excursion without being mistaken for the peaking mechanism.

A device-level implementation. The delivered netlist uses SKY130 MOS, poly-resistor and MIM models. External loading and common-mode excitation are explicitly documented testbench interfaces for the next stage of receiver integration.

05 / BIAS IMPLEMENTATION

Reference current made physical

The final export replaces the ideal Iref source and counts the bypass capacitor explicitly.

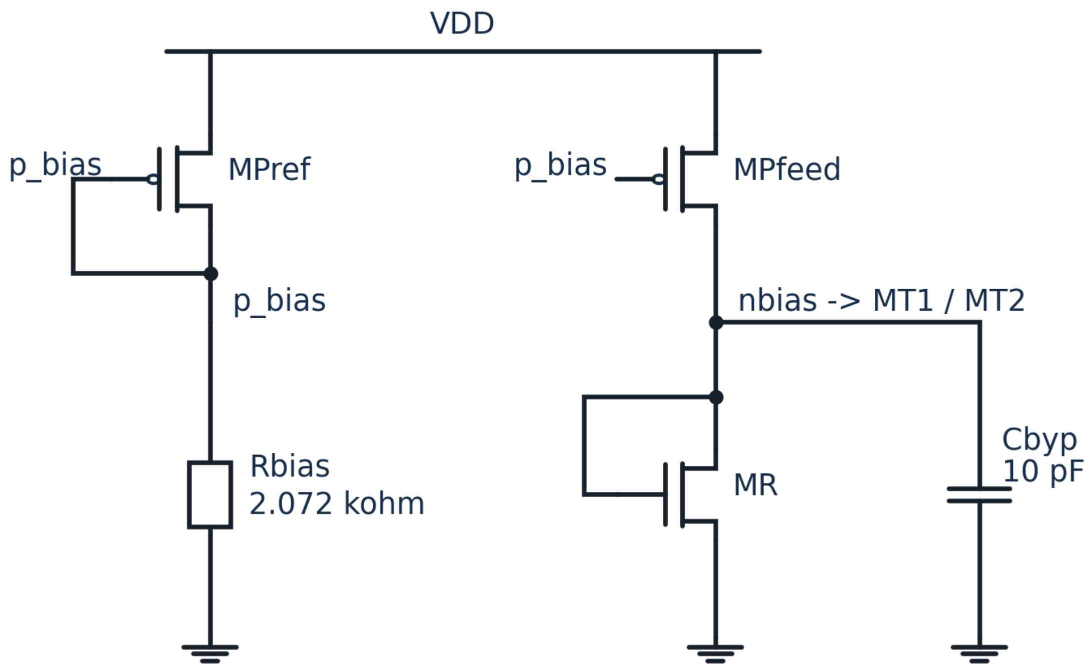

Figure 4. PMOS reference/feed mirror, poly bias resistor, diode-connected NMOS reference and MIM bypass. Equal net names connect to the CTLE. PMOS bulks = VDD; NMOS bulk = 0.

IMPLEMENTATION	VALUE / PURPOSE
MPref / MPfeed	W/L = 25.2069/0.5 um; PMOS supply-dependent reference
MR and tail mirror	MR W/L = 25.2069/0.5 um; each tail has 8x width
Rbias	Physical poly, W/L = 1/5.34 um; nominal 2.072 kohm
Cbyp	Physical MIM, 70.545 x 70.545 um; 9.99945 pF
Nominal supply power	6.966 mW, obtained from -VDD x I(VDD)

The sizing target sets the reference geometry; the operating current is then determined by the transistors and resistor. Supply-current measurement includes the reference branch and mirror error. The complete physical circuit is rechecked across PVT rather than assuming the nominal mirror ratio remains exact.

Design interpretation. This is a compact supply-dependent bias circuit, not a precision bandgap reference. Its practical value here is replacing an ideal source with a simulated physical implementation while preserving the fixed-circuit acceptance result.

06 / INPUT CONDITIONING

Physical attenuation and peaking

The input switches are real; the Rs/Cs search indices do not imply a fabricated tuning bank.

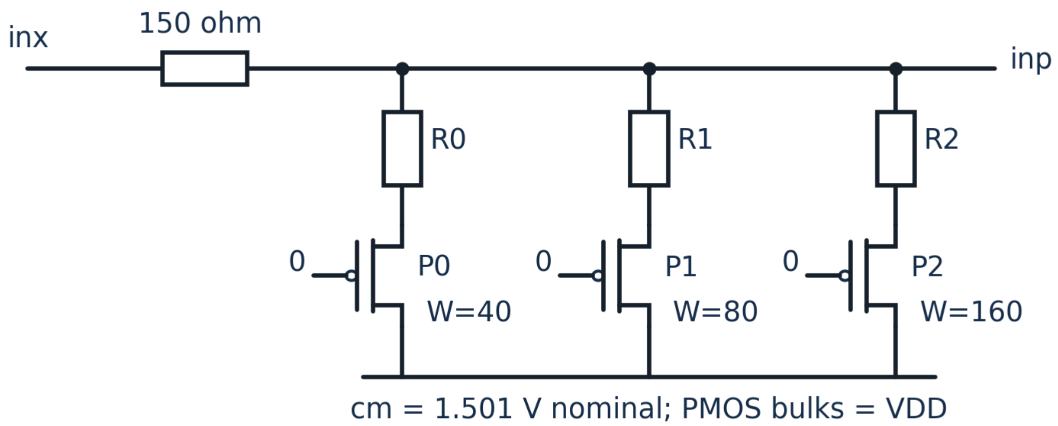

Figure 5. Positive input leg; the negative leg is identical. Three PMOS-switched shunt branches join inp to cm. A7 holds all three gates at 0 V (ON). W labels are total microns; L = 0.15 um.

The differential series-shunt network uses a 150 ohm series target per leg and binary-weighted shunt branches. Its switch resistance and parasitic loading are included in the measured response. In this export the attenuator code, Rs and Cs are held fixed across PVT.

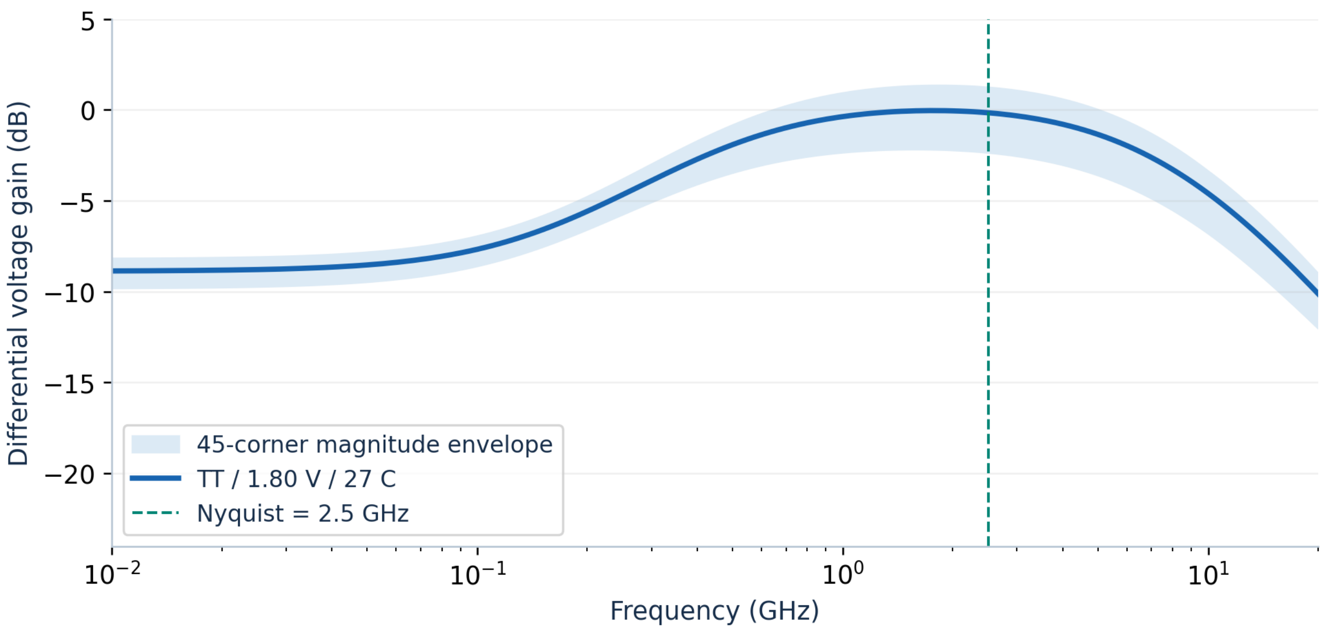

Figure 6. Raw SPICE AC magnitude: final nominal peaking 8.834 dB at 1.769 GHz; shaded limits show the 45-corner envelope. The curve is absolute voltage gain, not gain normalised to 0 dB.

Peaking is a ratio, not absolute gain. Peaking is the maximum AC gain relative to low-frequency gain. A CTLE can therefore provide about 9 dB of peaking while its absolute peak gain remains near 0 dB; channel insertion loss is a different quantity.

07 / FREQUENCY-DOMAIN ROBUSTNESS

One setting across the PVT grid

No corner-specific Rs/Cs retuning is used for this physical-product demonstration.

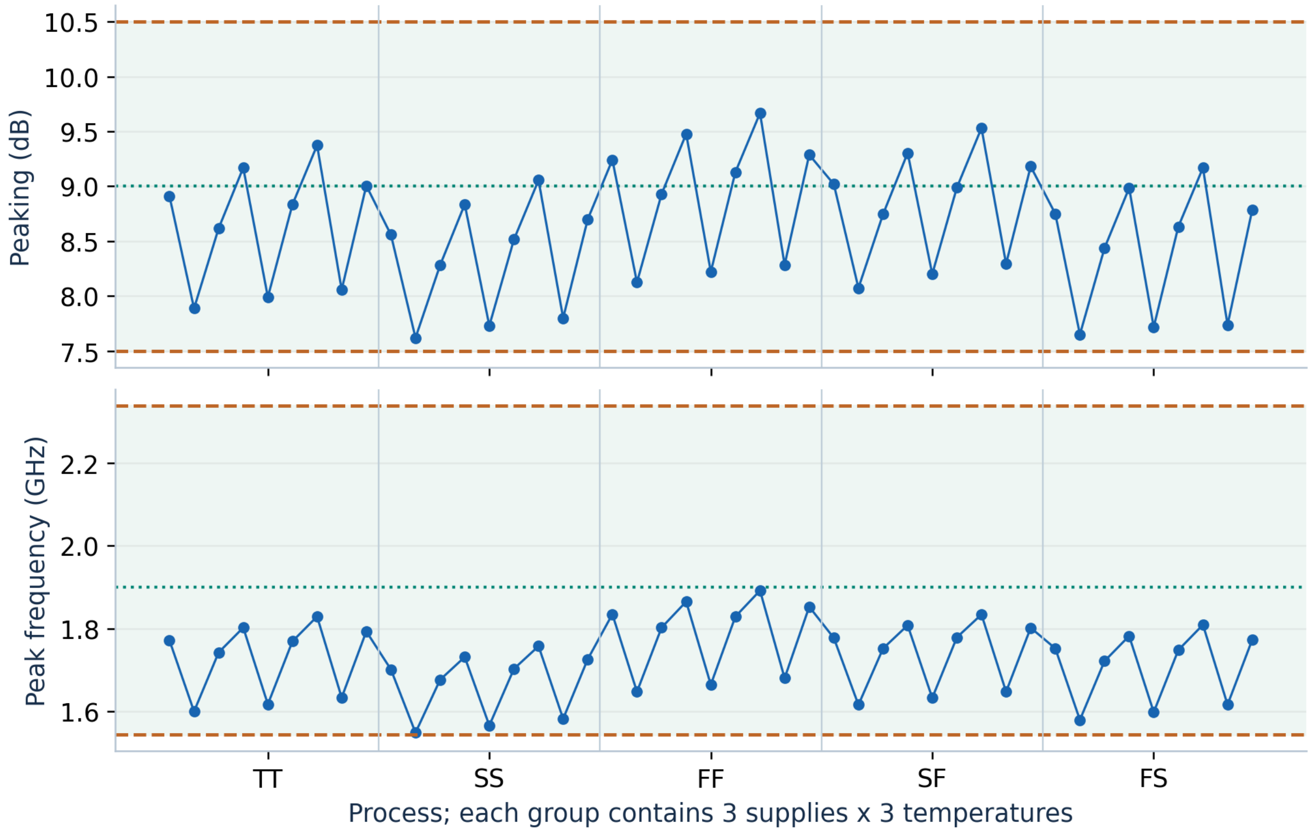

Figure 7. All 45 measured points. Shading and dashed lines mark project request-matching bands; dotted lines mark the requested centres. Point order within each process is supply/temperature order.

METRIC	NOMINAL	45-CORNER RANGE
Peaking	8.834 dB	7.619 to 9.667 dB
Peak frequency	1.769 GHz	1.548 to 1.892 GHz
Peak extraction	Log-frequency interpolation	Shared extraction rule for all points
Device-to-link fit residual	0.0860 dB	Maximum 0.1122 dB

The response shape follows the expected degeneration behaviour: approximately fz = 1/(2 pi Rs Cs), with degeneration pole multiplier k = 1 + (gm + gmb)Rs/2. The final response and peak are taken from SPICE, not these first-order equations.

Margin to watch. All points meet the declared response checks, but the tightest peak-frequency matching margin is 0.0044 octave. This is a sampled-corner result; mismatch, extracted parasitics and finer PVT sweeps remain separate sign-off steps.

08 / ELECTRICAL VERIFICATION

Noise and supply power

Results are measured at the differential interface of the final physical CTLE.

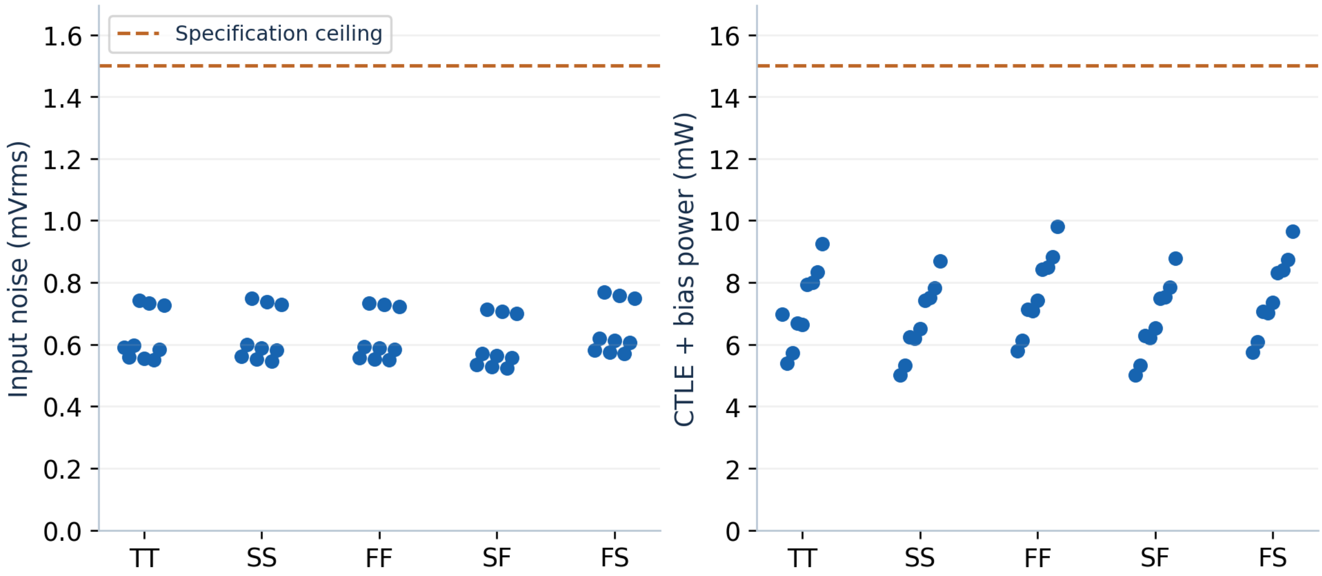

Figure 8. Nine operating points per process corner. Dashed lines are the competition ceilings; power includes the CTLE and physical bias reference, not a transistor DFE or clock network.

WORST OBSERVED	RESULT	REQUIREMENT
Input noise, 10 MHz-5 GHz	0.768 mVrms	< 1.5 mVrms
CTLE + bias power	9.802 mW	< 15 mW receiver target
HD3, 100 MHz	-81.49 dBc	< -30 dBc
HD3, 2.5 GHz	-47.08 dBc	Additional high-frequency check

Noise is referred to the differential input source before attenuation and integrated over the specified 10 MHz-5 GHz band. Supply power comes from the simulated VDD branch rather than nominal tail-current targets, so the physical reference and mirror behaviour are included.

Power excludes the decision circuit, feedback path, clocking, tuning controls and common-mode generator. HD3 is a simulated-model result: ngspice ignores the generic poly resistor voltage coefficients p2/q2/p3/q3. Their distortion contribution remains NOT_VERIFIED; see page 19.

Measurement discipline. The simulator returns integrated noise as RMS voltage, not a variance to square-root again. Output parsing, finite-value checks and transistor operating-point guards accompany every result; simulator exit status alone is not accepted as proof.

09 / LINEARITY AND HEADROOM

Validate the signal excursion

Harmonic distortion and DC swing complement small-signal AC measurements.

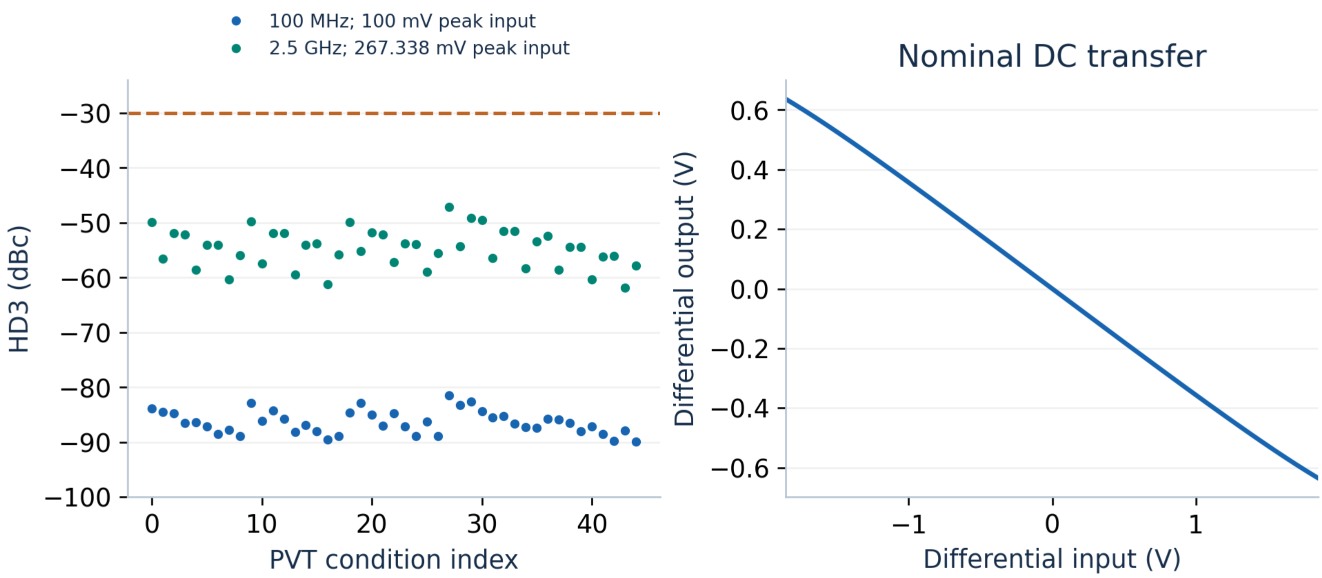

Figure 9. HD3 at two separately tested tone frequencies across PVT (left) and the raw nominal DC transfer (right). Input amplitudes are differential peak values, not peak-to-peak.

MEASUREMENT	TEST CONDITION / RESULT
Competition HD3 test	100 MHz, 100 mV differential peak; worst -81.49 dBc
Additional Nyquist test	2.5 GHz, 267.338 mV differential peak; worst -47.08 dBc
Nominal output swing guard	Measured limit 1.104 V differential peak-to-peak
Nominal saturation margins	Input pair 0.649 V; tail 0.350 V (VDS - VDSAT)

HD3 is computed as 20 log10(V3/V1) using a coherent steady-state window: 20 fundamental cycles and 2,000 uniform samples. The same extraction routine is used at both frequencies. The differential DC transfer is inspected separately because an acceptable small-signal response does not guarantee adequate large-signal headroom.

Device-to-link guard

The FFT results reproduce, but generic poly resistor voltage coefficients are ignored by ngspice. Full passive linearity remains NOT_VERIFIED. Replacing those resistors changes resistance and parasitics and requires fresh circuit verification; numerical HD3 margin cannot bound the omitted effect.

10 / SYSTEM BRIDGE

From transistor response to eye

Absolute signal amplitude is carried into a 5 Gbps NRZ link model.

The bridge fits a one-zero/two-pole response to the measured CTLE magnitude, checks fit quality and measured output swing, and composes it with the transmitter and channel. Pulse cursors then give worst-ISI opening after first-post-cursor cancellation. This links device choices to a receiver metric without a transient SPICE run for every bit pattern.

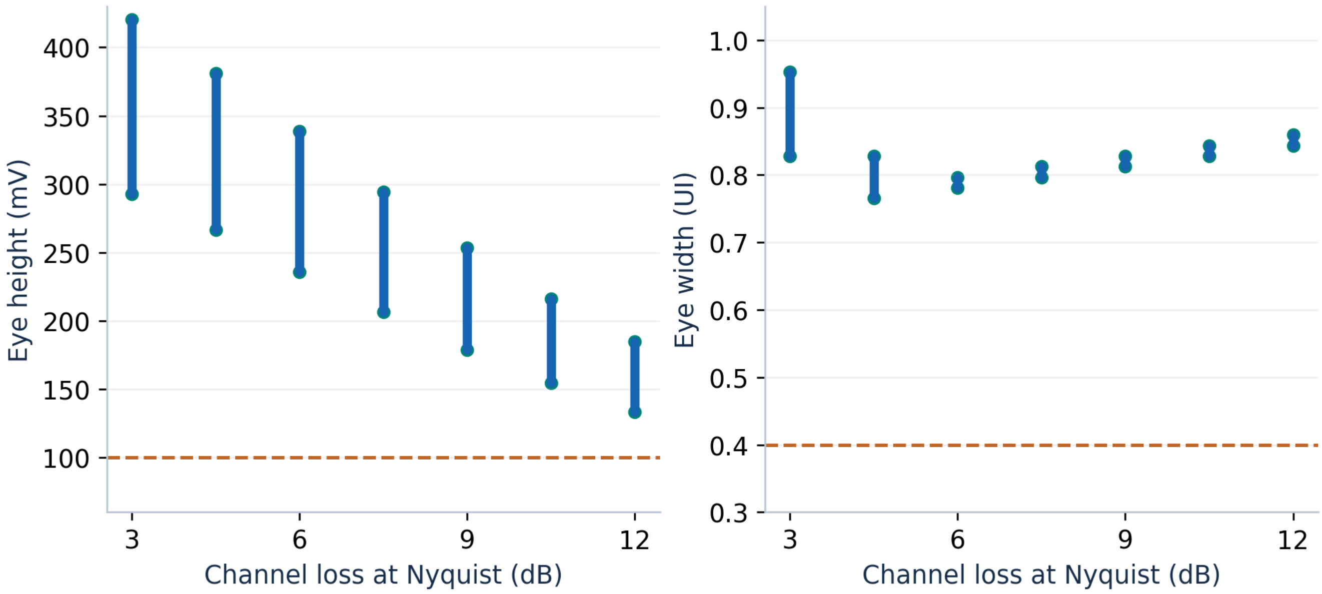

Figure 10. Minimum-to-maximum modelled eye over the 45 PVT points at each constructed channel loss, with ideal behavioural 1-tap DFE. Dashed lines: 100 mV and 0.4 UI thresholds.

LINK CONDITION	DECLARED SETTING
Transmitter	0.8 V differential peak-to-peak; -3.5 dB de-emphasis
Channel family	Constructed minimum-phase skin/dielectric-loss channels; 3 to 12 dB at Nyquist in 1.5 dB steps
Sampling / eye definition	64 samples/UI; height at pulse cursor; contiguous positive-opening width
Final circuit across all conditions	One fixed geometry and code; no channel-specific resizing
Lowest ideal-DFE opening	133.52 mV; 0.7656 UI (separate minima)

What this visual establishes. These are noiseless cursor-based ISI openings: noise is not subtracted and width is measured at zero height. They are not BER contours. Phase is inferred from the magnitude fit, not independently validated; random jitter, clock recovery and DFE error propagation are absent.

11 / DECISION FEEDBACK

What the 1-tap DFE contributes

A controlled behavioural model makes the cancellation assumption inspectable.

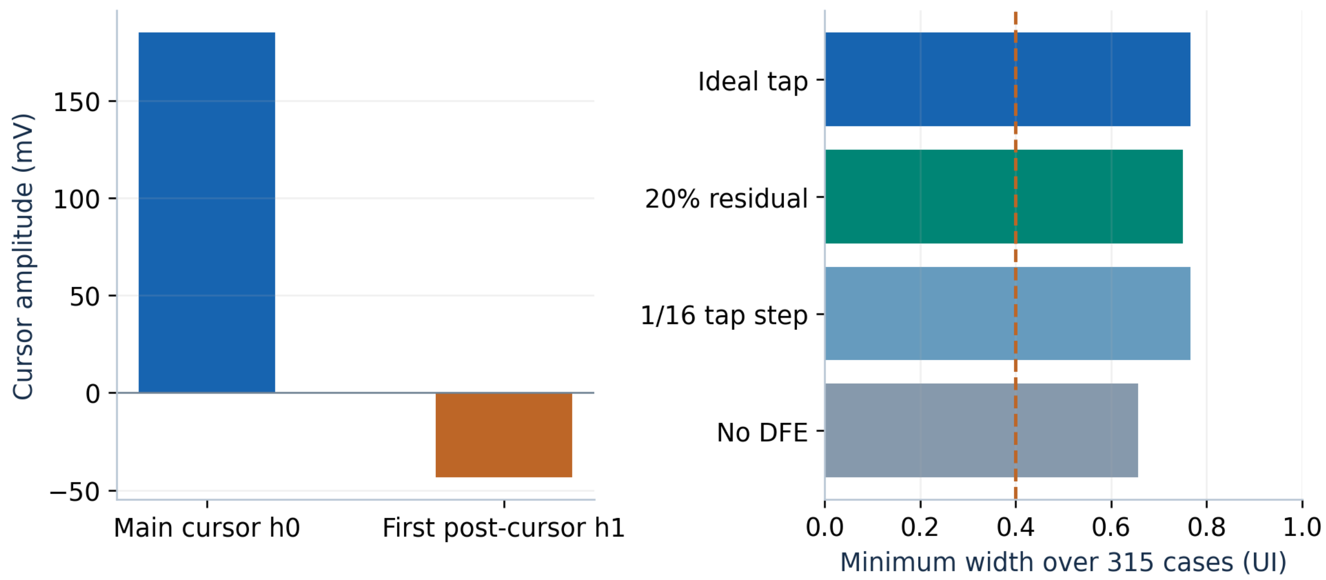

Figure 11. Nominal 7.5 dB-channel cursors (left) and worst width under four DFE policies over all 315 conditions (right). These are saved link-model results, not transistor-level DFE measurements.

Operation. The normalised tap is b1 = h1/h0. The previous decision removes the first post-cursor, while the remaining pre- and post-cursors set the worst-ISI opening. For ideal cancellation, eye height is 2(h0 - sum of the remaining absolute cursor amplitudes), clipped at zero.

TAP POLICY	MIN HEIGHT	MIN WIDTH
Ideal first-post-cursor cancellation	133.52 mV	0.7656 UI
20% of first post-cursor remains	133.21 mV	0.7500 UI
Tap step 1/16; clipped to +/-0.5	131.94 mV	0.7656 UI
No cancellation	131.94 mV	0.6562 UI

All four policies remain above the eye thresholds for this circuit and channel family. The clearest benefit here is width margin; the CTLE itself already supports the height target. The tap-quantisation model includes both endpoints and must not be presented as a realised 4-bit DAC.

For the nominal 7.5 dB channel, h0 = 184.99 mV and h1 = -43.43 mV, giving b1 = -0.2348. The ideal-tap opening is 255.81 mV and 0.8125 UI. This concrete operating point connects the cursor plot to the cancellation arithmetic.

Hardware boundary. The final DFE is behavioural. The latest physical decision/hold study passed 14/32 held bits; all 18 transitions missed the timing window and some terminals exceeded the model domain. The failed prototype is preserved in DFE_LOW_LOAD_RESULTS.md; no integrated transistor DFE is demonstrated.

12 / REINFORCEMENT LEARNING

Learning a useful search policy

The policy learns incremental improvement inside a bounded, characterised design space.

RL ELEMENT	IMPLEMENTATION
State	62 features: six request/current-search fields plus eight seven-field history slots. Each slot records presence, three indices, measurement validity, eye height and eye width. PVT/channel identity and oracle quality are hidden.
Actions	Increase/decrease attenuation, Rs or Cs index; or LOCK. Seven actions, with bounds masks.
Episode	At most eight visits; lock is available after two visits. After rollout, the shield selects the best compliant visited candidate, including the start.
Network / training	Two 64-unit hidden layers; imitation warm start followed by PPO; five independent training seeds.
Training scale	50 imitation epochs and 200,000 PPO steps per seed. Frozen policy is reused at inference.

Reward: improve feasible eye quality

Normalised quality and incremental reward

$q(s) = \mathrm{clip}(A(s)/A^*, 0, 1)$ for a compliant state; otherwise $q(s) = 0$.
Move reward = $q(\text{next}) - q(\text{current})$.
LOCK reward = 0 if compliant; otherwise $-1 - q(\text{current})$.
A is eye-opening area; A^* is the per-condition feasible reference. Discount factor: 0.99.

Reward and compliance are distinct: a large eye does not compensate for failing noise, power or response checks. The training oracle supplies quality labels; the deployed actor sees only its observation. The deterministic verifier remains the authority on acceptance.

The actor therefore learns from how candidate changes affect the observed eye. Action masks prevent out-of-range index moves, while the safety shield adds constraint knowledge outside the neural network. These roles are deliberately separate in the implementation.

Why this formulation. Early continuous-search experiments motivated a bounded library formulation with explicit requests and feasibility checks. This made improvement measurable, training reusable and the final decision explainable instead of relying on an opaque reward alone.

13 / EXPERIMENTAL EVIDENCE

The frozen hybrid-policy experiment

Original five-seed held-out result, preserved. Supplemental attribution controls appear on page 18.

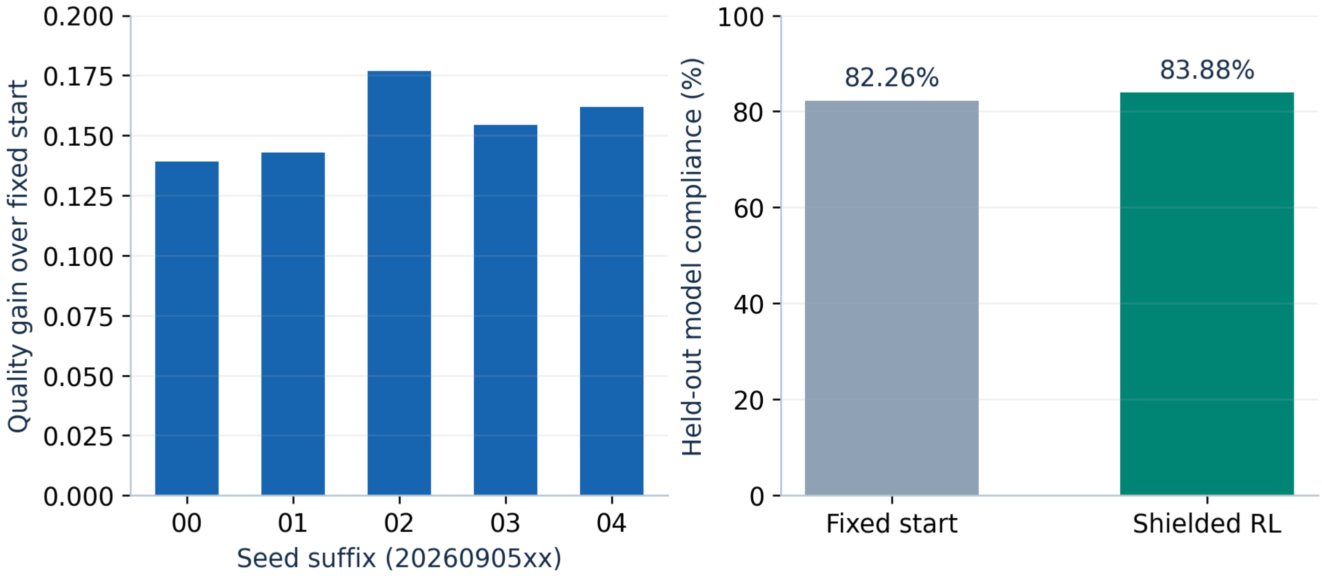

Figure 12. Every seed improves normalised quality over the fixed start. Compliance compares the same held-out conditions. This is the frozen library-policy study [E3], not a re-training result for the later physical-bias circuit.

HELD-OUT RESULT	VALUE
Condition identities	2,430 per seed; 12,150 seed-condition evaluations
Mean normalised quality	0.5619 fixed start -> 0.7169 shielded RL
Mean paired quality improvement	+0.1550; 95% bootstrap CI [0.1508, 0.1593]
Positive training seeds	5 of 5
Mean policy visits	5.579 per evaluated identity
Mean model compliance	82.26% -> 83.88%

The held-out set exercises midpoint loss/request conditions within the established device library and PVT lattice. The result supports a repeatable improvement in feasible eye quality. It is not evidence of generalisation to a new process, a measured board channel or an arbitrary circuit topology.

Interpretation for the submission. RL contributes useful, repeatable local improvement. The final physical product adds deterministic fixed-circuit selection and fresh verification on top; it should be presented as RL-assisted automation, not as proof that RL alone guarantees every specification.

14 / ENGINEERING JOURNEY

How the final architecture emerged

Each stage converted a modelling assumption into a more testable engineering decision.

01

Physics and a reproducible simulator interface

Established the SKY130 CTLE, differential units, source degeneration and device-to-link bridge. Added output parsing and operating-point guards so a simulator exit code was not the sole evidence.

02

A bounded, specification-driven search problem

Organised the candidate space as eight attenuation choices x eight Rs choices x eight Cs choices. Reused measured circuit data for policy training and request-specific evaluation.

03

A shielded policy with held-out evidence

Combined imitation warm start, PPO and best-compliant-visited selection. The original result improves quality over a fixed start; page 18 separately tests imitation and classical controls.

04

From adaptive candidates to one robust export

Added target-centred matching and all-corner intersection. Selected setting 490 so the physical demonstration uses the same circuit across every tested process, voltage and temperature.

05

Physical bias, bypass and auditable geometry

Replaced ideal Iref with a PMOS/resistor reference, realised the bypass as MIM and remeasured the whole candidate. Bound raw outputs, schematic and inventory to one circuit identity.

06

Product integration and a controlled submission scope

Integrated the physical mode into CLI and browser workflows. Hardware decision/retention studies failed transition timing and model-domain checks; the submission preserves the verified CTLE and behavioural DFE.

DESIGN DECISION	RESULTING EVIDENCE
Freeze the learning policy	5/5 seeds with positive held-out quality gain
Freeze one physical circuit	45 PVT points; 315 electrical-model cases
Bind the exported artifacts	616 raw/evidence hashes; exact nominal deck

The final design choice. The final architecture balances deployability, reproducibility and traceable device-level verification. Each stage strengthened the delivered workflow: from physical models to reusable learning, fixed-circuit acceptance and inspectable exports.

15 / PRODUCT AND EFFICIENCY

A usable engineering workspace

A structured request becomes a circuit artifact, not merely a reward curve.

PRODUCT CAPABILITY	ENGINEERING USE
Structured / natural-language input	Translate intent into peaking and peak-frequency targets; confirm the actual structured values.
PVT exploration	Inspect corner-dependent metrics and margins rather than only nominal headline values.
Circuit comparison	Compare selected circuit artifacts and their response; no benchmark leaderboard is required in the user workspace.
Channel upload	Supports a separate channel-assessment workflow. The 315-case result in this report uses the declared constructed family, not arbitrary uploaded data.
Failure-aware output	Expose unsupported requests and unverified scope without manufacturing a pass.
Evidence bundle and demonstration view	Export circuit, schematic, JSON, raw evidence and hashes. Cached legacy demo artifacts remain distinguishable from the final physical run.

Where the computational work happens

Offline: 512 settings x 45 PVT points = 23,040 measured library rows; seven channel views give 161,280 condition rows. Additional midpoint characterisation and five-seed training are separate development costs.
Online physical example: the saved final run used 137 fresh simulator calls: one legacy capture, one bias calibration and 135 PVT measurement calls. Recorded run wall time: 107.91 s on the development machine.

The physical mode uses legacy policy proposals and a complete fixed-setting prescreen before fresh acceptance. Its table accesses are not SPICE calls. The 5.579 mean policy visits belong to the held-out RL experiment, not the complete physical-export runtime.

Suggested demonstration sequence

Enter 9 dB and 1.9 GHz, select the physical-bias mode, then inspect nominal response and the fixed PVT matrix. Open the schematic and export the bundle. For an immediate review, use the saved physical example instead of starting a fresh run.

Efficiency at a defined boundary. Characterisation is amortised across requests and policy reuse reduces candidate visits in the reported experiment. Recorded costs distinguish training, table search and fresh verification. Original and midpoint banks plus five PPO runs total about 6.71 hours of recorded stage runtimes, excluding imitation and earlier development. End-to-end speedup remains unverified.

16 / AREA AND INTEGRATION

Geometry counted at the right boundary

A physical capacitor can dominate area even when MOS gate rectangles are small.

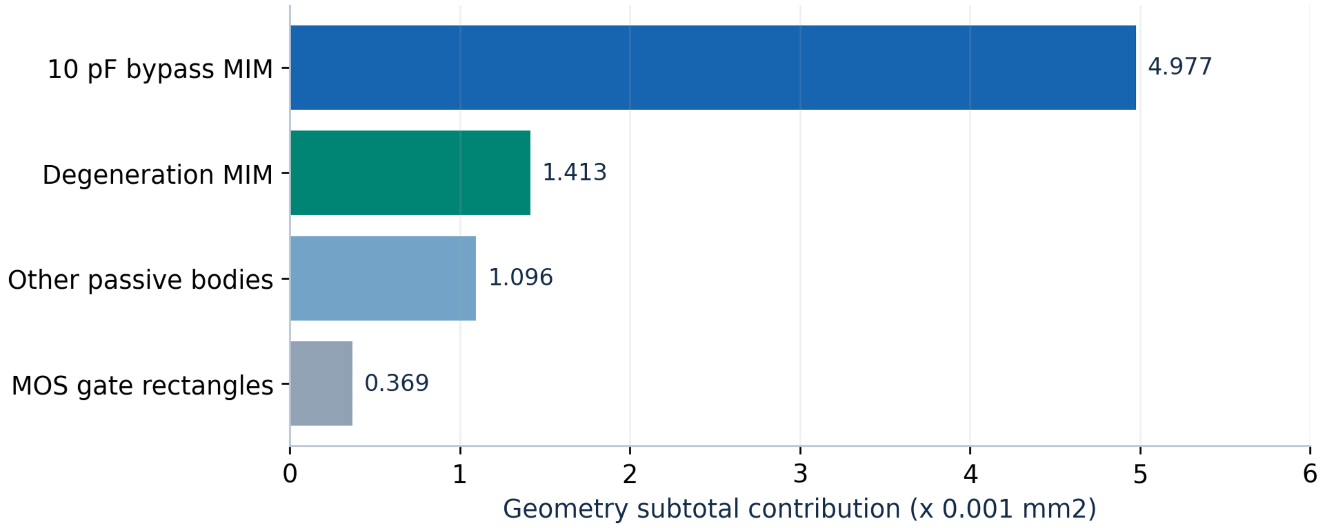

Figure 13. Netlist-derived geometry inventory for all 27 physical instances. Values count passive bodies/plates and MOS W x L rectangles, with instance multiplicity; they are not a routed floorplan.

AREA ACCOUNT	RESULT
Physical bypass MIM plate	0.004977 mm2
All passive bodies / plates	0.007486 mm2
All MOS gate rectangles	0.000369 mm2
Known geometry subtotal	0.007855 mm2
Complete receiver area vs 0.05 mm2	NOT_VERIFIED; no guessed layout multiplier

The inventory includes the input attenuator, core, physical reference, degeneration elements and bypass. Contacts, diffusion, wells, guard rings, routing, spacing and a floorplan are not represented by the subtotal. The transistor DFE, clocks, Rs/Cs selectors and common-mode generator also require an integration budget.

Physical Rs/Cs tuning is an extension to the present fixed-geometry export. Selector parasitics must be characterised in the final topology, followed by complete receiver simulation and layout extraction. The assumed 32.628 fF load per output must be replaced or justified when a real next stage is attached.

Next technical closure. Integrate the decision/feedback path and control, verify device operating limits and timing, then perform layout-aware area/power and post-extraction PVT checks. These steps extend the delivered platform; the current subtotal is not used to certify full S7 compliance.

17 / REPRODUCIBILITY

Evidence an evaluator can inspect

Every headline result is tied to saved source artifacts; report generation performs no new SPICE or RL run.

Reproduce the physical product

From the repository root, in the documented Python/ngspice/PDK environment:
py -3.13 -m nebula.design --method rl-physical --peaking 9 --f-peak 1.9 --out output/my_physical_design
Use a new output directory. The command launches fresh acceptance; opening the saved demonstration is the quick review path.

[E1] Final physical export. nebula/product_demo/physical_bias_9db_1p9ghz_20260906/
design.json, design.cir, physical_evidence/summary.json and fixed_pvt.jsonl; 616 evidence-file hashes checked.

[E2] Physical implementation and product integration. nebula/physical_design.py; nebula/PHYSICAL_PRODUCT_RESULTS.md
One fixed signature, physical reference/MIM generation, fresh acceptance and artifact export.

[E3] Frozen RL experiment. nebula/experiments/shielded_policy_final_results.json
Five seeds, paired held-out metrics and bootstrap confidence interval; policy freeze a84082e.

[E4] Reward and deterministic verification. nebula/rl/margin_improve_env.py; nebula/rl/safety_shield.py
Incremental quality reward, action validation and termination rules.

[E5] Device-to-eye and DFE controls. nebula/link/bridge.py; nebula/link/cursors.py; nebula/link/dfe_ablation.py
Measured-response adapter, cursor definitions and four cancellation policies.

[E6] Geometry and exact-deck checks. nebula/report/product_scope.py; nebula/report/physical_schematic.py
Physical inventory, ordered connectivity checks and canonical circuit identity.

Method and tool references

[R1] Schulman et al., Proximal Policy Optimization Algorithms, 2017. arxiv.org/abs/1707.06347
[R2] SkyWater, SKY130 device documentation. SKY130 device details
[R3] ngspice documentation. ngspice.sourceforge.io/docs.html
References accessed 6 September 2026. Installed simulator: ngspice 41.

A short evidence-review protocol

1. Match design.json to the nominal TT values and inspect the exported transistor/passive deck.
2. Confirm one circuit signature across the 45 PVT journal rows, each with seven link conditions.
3. Inspect raw AC, noise, HD3 and swing files for a nominal and a limiting corner.
4. Review the separate RL held-out result and geometry inventory at their stated boundaries.

Artifact integrity

Physical circuit signature: 21f09626c56aeb70... . The companion report source manifest records exact source and output SHA-256 hashes. The final report separates physical CTLE evidence, library-policy experiments and system-model assumptions throughout.

18 / POST-REVIEW ATTRIBUTION

What PPO adds, and what it trades

Supplemental exposed-data diagnostic, registered at 030b8f2; no training or new SPICE.

SAME 8-VISIT CAP	COMPLIANCE	QUALITY q	ACTUAL VISITS
Fixed start	82.26%	0.5619	1.000
Visible-eye hill climb	84.07%	0.6339	8.000
Random local moves	90.79%	0.6508	8.000
Uniform global samples	88.08%	0.6273	8.000
Imitation only	82.35%	0.6769	4.474
Imitation + PPO	83.88%	0.7169	5.579

All arms use the same starting lookup and best-compliant-visited shield. Random arms use 20 persistent seeded streams; imitation and PPO use five matching frozen checkpoints. Local controls share one-index moves; global sampling has a different move space and is a practical sizing control. Repeats are billed.

PPO versus imitation: quality +0.0400, paired 95% interval [0.0168, 0.0644]; compliance +1.53 percentage points. PPO uses more visits than imitation. Random local exploration achieves higher compliance, while PPO achieves higher mean quality with fewer visits.

EXTERNAL VISIT CAP	IMITATION q	PPO q	RANDOM LOCAL q
2	0.6078	0.6130	0.5977
4	0.6667	0.6764	0.6257
8	0.6769	0.7169	0.6508

Caps interrupt original eight-visit trajectories; trained observation scaling stays unchanged and early LOCK is honoured. Intervals resample 54 request/loss blocks after averaging seeds and PVT within each block. They are conditional on the shared circuit library. Every eight-visit PPO output reproduces the original result.

Source: nebula/product_audits/entry113_attribution_20260907/summary.json and episodes.jsonl.gz. The earlier FINAL is now exposed; this supplement is not a new held-out evaluation. Cached CPU timings are not SPICE speedups.

Defensible contribution. PPO adds measurable quality and compliance over imitation at the same maximum budget. Against classical controls the evidence shows a quality/compliance trade-off, not universal superiority. The final physical selection is a separate classical-intersection step.

19 / PHYSICAL COVERAGE AND MODEL AUDIT

The demonstrated operating boundary

Twelve predeclared requests through the unchanged physical product; all refusals and failures retained.

TARGET dB / GHz	FIXED SETTING	MODEL CASES	OUTCOME
3 / 1.25	--	--	BANK NO FIXED
3 / 1.9	273	312/315	MODEL FAIL
3 / 2.5	--	--	BANK NO FIXED
6 / 1.25	--	--	BANK NO FIXED
6 / 1.9	288	314/315	MODEL FAIL
6 / 2.5	480	308/315	MODEL FAIL
9 / 1.25	--	--	BANK NO FIXED
9 / 1.9	490	315/315	MODEL PASS
9 / 2.5	--	--	BANK NO FIXED
12 / 1.25	--	--	BANK NO FIXED
12 / 1.9	--	--	BANK NO FIXED
12 / 2.5	--	--	BANK NO FIXED

1/12 requests pass the sampled electrical-model gate; 8/12 have no fixed candidate in the characterised legacy bank. The new matrix used 548 fresh SPICE calls. Each measured request uses fixed geometry across 45 corners. This is circuit generation, not physical switching.

BANK NO FIXED denotes bank refusal; MODEL FAIL denotes failure of the measured candidate. coverage.jsonl separately records request matching and absolute response bounds. Full receiver area, power and transistor DFE remain unverified.

Model audit: a warning with a physical consequence

Raw logs confirm ignored p2/q2/p3/q3 in generic res_high_po. HD3 passes for that model; omitted resistor voltage dependence remains NOT_VERIFIED. Fixed-width families support voltage expressions but change resistance and parasitics; equivalent replacement is unverified.

Sources: nebula/product_audits/entry113_coverage_20260907/ and entry113_models_20260907/. Both include hashes. No reward, range, policy or circuit-topology changes were made. The original physical demonstration and frozen RL results are preserved.

Submission scope. This evidence supports automatic generation and verification within a limited characterised domain. It does not establish the full tuning envelope, complete receiver compliance, or an RL advantage in choosing the final physical circuit.